

ATQASUK, AK, USA

WMO: 702685

Lat: 70.467N

Lon: 157.436W

Elev: 96

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 27518

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.7	-37.7	-35.7	0.7	-28.4	-34.8	0.8	-28.1	39.9	-1.1	35.9	-3.4	4.6	80	1.057

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	67.6	57.3	63.4	55.1	59.5	53.1	58.0	65.5	55.6	62.1	53.5	58.8	12.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.0	62.3	61.4	52.0	57.8	58.7	50.2	53.9	56.8	25.1	65.4	23.6	62.3	22.3	58.8	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.9	28.0	25.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	-14.2	-11.6	-12.3	3.6	23.9	42.2	50.6	43.7	34.9	21.8	3.0	-8.0
	DBStd	25.69	14.32	16.15	13.74	11.72	10.21	8.05	6.57	6.56	6.62	7.16	11.57	13.37
	HDD50	12935	1991	1725	1933	1392	810	257	74	218	454	873	1410	1798
	HDD65	18271	2456	2145	2398	1842	1275	683	447	660	904	1338	1860	2263
	CDD50	139	0	0	0	0	0	24	92	23	0	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	11	0	0	0	0	0	1	10	0	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	12.1	12.6	12.6	11.5	12.8	12.4	13.0	11.9	11.7	11.7	12.9	11.2	11.5
	PrecAvg	9.7	0.5	0.5	0.4	0.3	0.4	0.6	1.3	1.9	1.4	1.2	0.7	0.5
Precipitation	PrecMax	18.8	1.7	1.7	3.8	1.1	1.3	2.1	2.6	3.7	5.2	2.8	2.4	2.2
	PrecMin	4.9	0.1	0.1	0.0	0.0	0.1	0.0	0.4	0.6	0.4	0.5	0.2	0.1
	PrecStd	3.3	0.4	0.4	0.7	0.2	0.3	0.4	0.6	0.9	1.0	0.6	0.4	0.5
	PrecStd	3.3	0.4	0.4	0.7	0.2	0.3	0.4	0.6	0.9	1.0	0.6	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	29.9	29.6	30.9	33.5	46.1	70.9	74.5	68.4	58.8	39.2	32.4	29.3	28.4
	0.4% MCWB	29.2	28.7	29.5	31.7	42.4	58.5	60.4	57.2	51.3	37.2	31.8	28.6	28.4
	2% DB	22.8	24.2	22.7	27.8	42.2	64.2	69.3	62.5	52.9	36.0	28.1	24.7	24.7
	2% MCWB	22.3	23.6	21.9	26.7	38.9	54.0	58.3	54.6	47.7	34.8	27.7	23.8	23.8
	5% DB	12.2	19.0	15.5	24.5	38.9	59.7	65.4	58.1	48.3	33.3	23.2	18.2	18.2
	5% MCWB	11.6	18.5	15.0	23.8	36.5	51.7	56.5	52.5	45.7	32.6	22.5	17.8	17.8
	10% DB	5.2	13.5	6.1	20.1	36.2	55.3	61.8	54.4	44.7	30.9	19.4	10.5	10.5
	10% MCWB	4.9	12.8	5.3	19.7	34.4	49.4	54.9	50.5	42.8	30.3	19.1	9.8	9.8
	0.4% WB	30.0	28.9	30.2	31.8	42.8	59.1	62.5	57.8	51.4	37.7	32.0	28.4	28.4
	0.4% MCDB	30.7	29.6	31.6	33.4	45.6	69.9	73.4	65.5	57.0	38.9	32.6	29.0	29.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	22.8	24.3	22.5	26.9	39.2	54.8	59.4	55.3	48.3	34.9	28.0	24.0	24.0
	2% MCDB	23.5	24.9	23.3	27.7	41.8	62.9	67.5	60.7	52.4	35.7	28.2	24.9	24.9
	5% WB	13.4	19.5	15.8	23.8	36.5	52.1	57.3	53.0	45.9	32.7	22.6	18.2	18.2
	5% MCDB	13.8	20.0	16.3	24.5	38.7	58.7	63.6	57.4	48.4	33.4	23.2	18.5	18.5
	10% WB	6.6	14.4	6.9	19.7	34.4	49.7	55.2	50.9	42.6	30.3	19.0	10.4	10.4
	10% MCDB	7.1	14.9	7.8	20.1	36.0	55.4	61.4	54.3	44.6	30.8	19.4	11.0	11.0
Mean Daily Temperature Range	MDBR	12.5	13.2	12.5	13.0	9.0	13.6	14.3	10.9	8.1	8.5	11.3	12.2	12.2
	5% DB	20.3	18.2	15.6	14.8	11.7	20.3	20.3	18.1	15.6	9.0	13.5	16.2	16.2
	5% WB	20.1	18.1	15.4	14.3	8.8	12.3	11.1	11.2	10.8	8.0	13.3	15.9	15.9
	MCDBR	20.6	18.1	15.5	14.7	11.5	19.4	18.8	16.3	15.5	8.7	13.7	16.2	16.2
	MCWBR	20.4	17.9	15.3	14.2	8.9	12.4	10.9	10.8	10.9	8.0	13.4	16.0	16.0
	MCWBR	20.4	17.9	15.3	14.2	8.9	12.4	10.9	10.8	10.9	8.0	13.4	16.0	16.0
Clear-Sky Solar Irradiance	taub	N/A	0.208	0.244	0.305	0.324	0.327	0.338	0.325	0.286	0.208	N/A	N/A	N/A
	taud	N/A	2.043	2.235	2.129	2.250	2.501	2.497	2.534	2.545	2.145	N/A	N/A	N/A
	Ebn at Noon	0	163	247	259	270	275	267	257	237	161	0	0	0
	Edh at Noon	0	12	21	33	34	28	27	23	16	10	0	0	0
All-Sky Solar Radiation	RadAvg	0	102	579	1302	1735	1794	1521	935	461	130	0	0	0
	RadStd	0	6	24	81	110	143	164	91	43	14	0	0	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	+2.63	+4.82	+2.48	N/A	N/A	N/A	-869	-873	N/A	N/A	

Nomenclature: See separate page